American Sign Language Recognition Using Convolutional Neural Networks

COMP 472 Group J Concordia University Winter 2026

Problem Statement & Application • American Sign Language (ASL) is used by approximately 500,000 deaf and hard-of-hearing individuals in North America [1]; yet, communication barriers still limit their access to education, healthcare, and the workplace. Automated ASL recognition systems can reduce this gap through real-time translation and by encouraging more inclusive communication.

This project focuses on recognizing ASL hand gestures using Convolutional Neural Networks (CNNs). The first step is to classify three commands: *SPACE*, *NOTHING*, and *DELETE*. This is followed by the classification of digits 0–9, and finally the English alphabet. Throughout development, this project aims to improve classifier accuracy, evaluate performance on datasets of varying complexity, and gain experience with image classification techniques.

Previous CNN ASL recognition studies have faced challenges due to variations in lighting, background, and hand orientation; these can be addressed by using diverse datasets and data augmentation techniques [2]. However, visually similar signs (e.g., *M*, *N*, and *T*) remain difficult to distinguish due to overlapping finger positions [3].

The goal of this project is to develop and compare three CNN-based ASL recognition models, including models trained with transfer learning, and to see how they can improve communication for people who are hard-of-hearing.

Dataset Selection • Table 1 summarizes the key characteristics of the three datasets used in this study. It also presents representative sample images from each dataset to illustrate the visual appearance of the different categories of gestures.

Table 1. Summary of Selected ASL Image Datasets with Example Samples

Sample	Dataset	Classes	Num. Classes	Images/Class	Total Images	Resolution
	ASL Alphabets [4]	A–Z	26	1.5k	39k	200 × 200
	ASL Commands [4]	SPACE, DELETE, NOTHING	3	∼3k	∼9k	200 × 200
	ASL Digits [5]	0–9	10	218	2,180	100 × 100

*Subsampled ASL Alphabets from 87,000 to meet the 50,000 image constraint (∼1,500 images/class).

Proposed Methodology • The problem of recognizing a subset of ASL gestures from images will be solved using Convolutional Neural Networks (CNNs). This study aims to compare the performance of different CNN architectures on three datasets with different levels of classification complexity (commands, digits, and alphabet) to assess how dataset characteristics and architecture choice affect recognition accuracy and generalization. The classification complexity is influenced by the semantic similarity of gestures along with the number of gesture classes and the number of images available per class.

All datasets will be preprocessed using the same pipeline. Images will be resized to a fixed resolution of 224 x 224 pixels and normalized to the range $[0, 1]$. Data augmentation, such as rotation, flipping, and zooming, will be applied during training to help the models learn underlying patterns [6]. Each dataset will be split into 80% training, 10% validation, and 10% testing sets using stratified sampling to maintain class distribution.

The CNN architectures that will be used are MobileNetV2 (lightweight) [7], ResNet-10 (medium complexity) [8], and VGG16 (high complexity) [9]. They are configured to accept input images of size 224 x 224 pixels. Training these architectures on the three datasets will result in nine models. Additionally, two transfer learning models will be trained using pre-trained weights on selected dataset–architecture combinations. All models will be trained using categorical cross-entropy loss, the Adam optimizer, and early stopping to prevent overfitting. Google Colaboratory will be used to train and evaluate the models eliminating the dependency on local computational resources.

Performance metrics will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Comparisons will be made across datasets, architectures, and training strategies. It is expected that the low-complexity commands dataset will achieve the highest accuracy, whereas the alphabet dataset will be the most challenging. Lightweight models are expected to perform efficiently with lower computational cost, while deeper models may achieve higher accuracy at the expense of increased computational cost. The results of this project aim to help researchers in selecting appropriate CNN architectures and training strategies when developing ASL recognition systems.

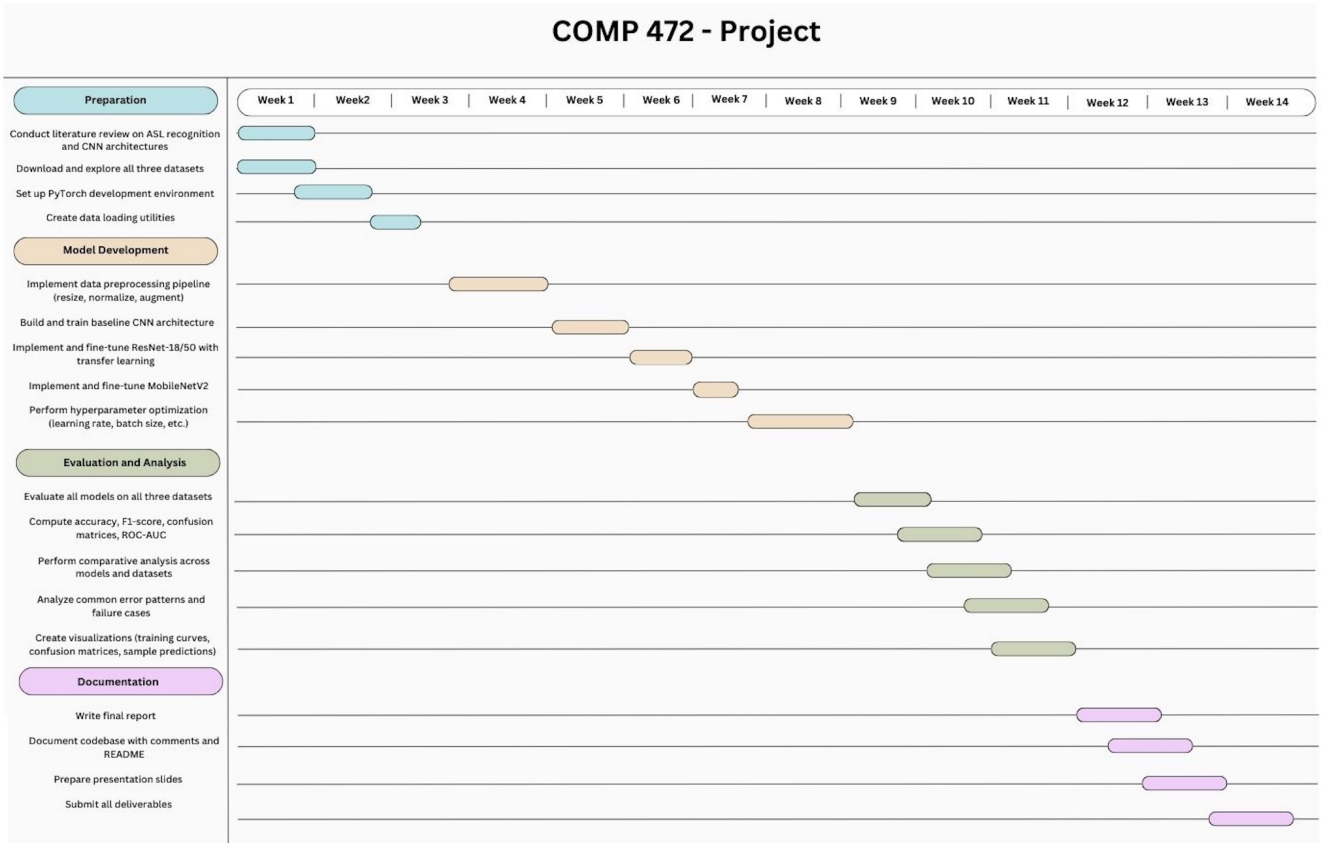

Figure 1. Project timeline (Gantt chart).

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